# Burkitt lymphoma

**MYC-driven, Ki-67 ~100%, curable** — requires TLS prevention, CNS prophylaxis and prompt intensive rituximab-containing therapy. Regimen choice depends on **age, fitness, and whether the CSF is involved**

*Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — Burkitt and HIV-associated lymphoma · NCCN · primary trial reports verified against PubMed, September 2026.*

## BIOLOGY AND DIAGNOSIS

| | |
|---|---|
| **Defining lesion** | **MYC rearrangement to an Ig locus**, confirmed by FISH: **t(8;14) IGH** *~80%* · **t(2;8) IGK** *~15%* · **t(8;22) IGL** *~5%*. Co-operating ID3, TCF3, TP53 mutations |
| **Morphology** | Monomorphic medium cells, diffuse. **"Starry sky"** — tingible-body macrophages engulfing apoptotic debris |
| **Phenotype** | CD20+, **CD10+, BCL6+, BCL2 negative**, TdT negative · **Ki-67 ~100%** |
| **Endemic** | Equatorial Africa, children; **EBV+ in nearly all**, malaria co-factor. **Jaw / orbit**, abdominal viscera |
| **Sporadic** | Worldwide, all ages; EBV+ in a minority. **Ileocecal / abdominal mass**, marrow, CNS |
| **Immunodeficiency** | **HIV** (often at relatively preserved CD4), post-transplant. Nodal, marrow and CNS disease common |
| **Not Burkitt** | **BCL2 or BCL6 rearranged too** → HGBCL double-hit (treat per DLBCL card). **MYC-negative with 11q gain/loss** → HGBCL with 11q aberration (WHO-HAEM5; ICC: LBCL with 11q), formerly "Burkitt-like". **TdT+** → lymphoblastic |

WHO-HAEM5 now subtypes Burkitt by **EBV status** (EBV+ vs EBV–) rather than by the epidemiologic variant.

## WORKUP

- **Excisional or core biopsy with MYC FISH** (plus BCL2, BCL6 FISH); start therapy within days
- **TLS labs at presentation:** K, phos, Ca, uric acid, creatinine, **LDH** — spontaneous TLS before any chemo is classic
- **PET/CT** · **bone marrow biopsy** · **LP with cytology AND flow cytometry** (brain MRI if symptoms) — CNS involved in **~19%** of adults
- **HIV**, HBsAg / anti-HBc / anti-HBs, HCV · **echo** · **G6PD before rasburicase** · pregnancy test · fertility counseling
- **No LP if circulating blasts or platelets <50,000**

## RISK STRATIFICATION

| | |
|---|---|
| **Low risk (regimen gate)** | **All of:** normal LDH, ECOG 0–1, stage I–II, no mass ≥10 cm, ≤1 extranodal site (Magrath / Mead). **NCI 9177 used mass <7 cm**, not 10. **Everyone else is high risk** (~85–90% of adults) |
| **BL-IPI (prognosis)** | 1 point each: **age ≥40** · **ECOG ≥2** · **LDH >3× ULN** · **CNS involvement**. 3-yr PFS **0 = 92%** · 1 = 72% · **≥2 = 53%** *Olszewski, JCO 2021* |

Real-world (30 US centres, 641 adults): 3-yr PFS **64%**, **treatment-related mortality 10%** — well below trial results. Rituximab (PFS 67% vs 38%) and **academic-centre care** (67% vs 46%) tracked with better outcome *Evens, Blood 2021*.

## TREATMENT PATHWAY

**Day 0 — before chemotherapy**

**TLS prophylaxis + staging in parallel**
IV fluids, **rasburicase**, labs q6–8h, strict I/O
LP / marrow; **always add rituximab** (LMB: EFS 75 vs 62%)

**Choose a regimen**

**Low risk**
**R-CODOX-M ×3** (A-A-A), or
**DA-EPOCH-R ×3**, no intrathecal (NCI 9177)

**High risk, CNS negative**
**R-CODOX-M / R-IVAC ×4** (A-B-A-B) if fit, <~65
**DA-EPOCH-R ×6 + IT MTX** if older, HIV+, comorbid — HOVON: similar PFS, less toxicity
R-hyper-CVAD / MA ×6–8 is an alternative

**High risk, CNS POSITIVE**
**HD-MTX + HD-cytarabine regimen** — R-CODOX-M/R-IVAC (16 IT total) or R-hyper-CVAD
**avoid DA-EPOCH-R**: CSF+ 4-yr EFS 45.5%

**Response**

**End-of-treatment PET/CT**
no maintenance, no consolidative ASCT in CR1
relapses are early (mostly <1 yr) and CNS-prone

**Relapsed / refractory**

**Clinical trial first**
salvage (R-ICE, R-DHAP) → ASCT if chemosensitive
allo-HCT for selected young, fit patients
**CD19 CAR-T off-label** — Burkitt not in the pivotal trials; bridge to transplant in case series

**Burkitt is the highest TLS-risk lymphoma** (Cairo-Coiffier high-risk category). **Rasburicase prophylaxis**, not allopurinol alone: label 0.2 mg/kg IV daily up to 5 days, though a **single fixed 6 mg dose** usually suffices (handbook: give for uric acid >7.5, renal dysfunction, or WBC ≥100,000). **Contraindicated in G6PD deficiency** (hemolysis, methemoglobinemia); post-dose uric acid in a **pre-chilled heparin tube on ice**. NS **2–3 L/m²/day**, goal UOP ~100 mL/h; **labs q6–8h**; continue **≥72 h after chemo**. **Do NOT alkalinize urine for TLS** — it promotes calcium-phosphate and xanthine precipitation (guideline: not recommended). Early nephrology if K, phos or creatinine climb. Allopurinol 300 mg BID after rasburicase.

## REGIMEN CHOICE — EVIDENCE

| Regimen | Evidence | Weak spot |
|---|---|---|
| **R-CODOX-M / R-IVAC** *modified Magrath* | LY10 (Mead 2008, MTX cut to 3 g/m²): 2-yr PFS **85% low risk, 49% high risk**, with less toxicity than full-dose. **Lowest CNS relapse** in real-world data (~4%) | Most toxic: grade 3–5 infection 56% (HOVON). Inpatient; age >65 needs dose reduction |
| **DA-EPOCH-R** | NCI 9177 (single arm, n=113, 25% HIV+): 4-yr **EFS 84.5%** — 100% low risk, 82% high risk; same across age and HIV. **HOVON/SAKK randomized** (CNS-negative high risk): PFS not significantly different vs R-CODOX-M/R-IVAC, fewer infections | **CNS.** CSF+ high-risk EFS **45.5% vs 89.9%**. Real-world CNS relapse **13% vs 3–4%** with MTX/cytarabine regimens (aSHR 4.38) |
| **R-hyper-CVAD / MA** | MDACC, n=31 (29% ≥60 yr): CR 86%, 3-yr OS **89%**, EFS 80%. Single arm | Long (6–8 cycles), myelosuppressive; small series |
| **Rituximab** | **Ribrag, Lancet 2016** (LMB ± R, n=260, randomized): 3-yr EFS **75% vs 62%**, HR 0.59, no added toxicity | — add to every regimen |

No regimen has beaten another for survival in a completed randomized trial. Practical rule: **fit, younger, or CNS-positive → HD-MTX/cytarabine-based**; **older, HIV+, or frail and CNS-negative → DA-EPOCH-R**.

## CNS PROPHYLAXIS

- **Prophylaxis is universal** (IT MTX ± cytarabine built into every regimen, + systemic HD-MTX in Magrath and hyper-CVAD). The one exception: **NCI 9177 low-risk patients received none** and none relapsed
- **CSF involvement is the strongest adverse factor** in every series — stage it properly (cytology + flow) before cycle 1 IT
- **CNS-positive:** intensify IT (Magrath: 2 extra per cycle, 16 total; hyper-CVAD: IT twice weekly until CSF clear ×2) — systemic HD-MTX and HD-cytarabine penetrate the parenchyma, infusional EPOCH does not

## HIV-ASSOCIATED BURKITT

- **Treat with curative intent, same regimens** — HIV status no longer predicts outcome (NCI 9177 EFS 84.9% HIV+ vs 84.5% HIV–; US-UK series 3-yr PFS 61%)
- **Continue or start ART** with oncology-pharmacy review; **avoid ritonavir / cobicistat boosting** (CYP3A4 → vincristine neuropathy, anthracycline and etoposide toxicity) — an integrase-inhibitor backbone is usual
- US-UK HIV-BL cohort (n=249, retrospective): **CNS recurrence higher after DA-EPOCH** (sHR 2.52), CODOX-M/IVAC longer PFS (aHR 0.45)
- PJP prophylaxis throughout; **MAC prophylaxis if CD4 <100**; consider quinolone during nadir

# Burkitt lymphoma



## MODIFIED MAGRATH (MEAD, BLOOD 2008) — Q21D OR ON COUNT RECOVERY

| Cycle | Dosing (mg/m² unless stated) |
|---|---|
| **A — R-CODOX-M** | Rituximab 375 d1 · cyclophosphamide **800 d1 + 200 d2–5** (first A; later A cycles 800 d1+2) · doxorubicin 40 d1 · vincristine 1.5 (cap 2 mg) d1+8 · **MTX 3000 over 24 h on d10** (300 over 1 h, then 2700 over 23 h) · IT cytarabine 70 mg d1+3 · IT MTX 12 mg d15 |
| **B — R-IVAC** | Rituximab 375 d1 · etoposide 60 d1–5 · ifosfamide 1500 d1–5 + mesna · **cytarabine 2000 q12h ×4** d1–2 (steroid eye drops) · IT MTX 12 mg d5 |
| **Age >65** | MTX **1000** (100 over 1 h, then 900 over 23 h) · ifosfamide 1000 · cytarabine **1000** |
| **CNS positive** | A: + IT cytarabine 70 mg d5, IT MTX 12 mg d17 · B: + IT cytarabine 70 mg d7 and d9 — **16 IT total** |
| **Leucovorin** | 15 mg/m² IV at **36 h** from MTX start, q3h to 48 h, then q6h until MTX <0.05 µM. After each IT MTX: 15 mg PO at 24 h |
| **Start cycle if** | ANC >1000, plt >75k, direct bili <1.4, Cr <1.2 — **not required for cycle 1 or for d10 HD-MTX** |

Rituximab was not part of LY10 (added in routine practice; Ribrag). G-CSF after the MTX level is <0.02 (A); d7–16 or pegfilgrastim d7 (B). Levofloxacin during nadir.

## DA-EPOCH-R (ROSCHEWSKI, JCO 2020) — Q21D

| | Dosing |
|---|---|
| **Drugs** | Rituximab 375 d1 · **etoposide 50, doxorubicin 10, vincristine 0.4 (no cap)**, each /day CIV d1–4 · cyclophosphamide 750 d5 · prednisone 60 BID PO d1–5 (handbook lists 60/day) |
| **Cycles** | Low risk **×3, no IT** · high risk **×6 + IT MTX 12 mg** each cycle (extended IT if CSF+) |
| **Adjust** | CBC twice weekly; ANC nadir ≥500 → **+20%** (E, D, C) · <500 on 1–2 checks → same · <500 on 3 checks or plt <25k → **–20%** |

G-CSF from 24 h after chemo. Outpatient-feasible via ambulatory pump.

## SUPPORTIVE CARE THAT TRIPS PEOPLE UP

- **Hold TMP-SMX around HD-MTX** (delays clearance) — use atovaquone or pentamidine for PJP, or resume once MTX has cleared
- **Avoid azoles that inhibit CYP3A4 (posaconazole, voriconazole, itraconazole) with vincristine** — severe neuropathy and ileus. Bowel regimen with every vincristine dose
- **HD-cytarabine:** prednisolone 1% drops 2 gtt OU BID from 12 h before to 48 h after; **check for cerebellar signs before each dose** (higher risk with age >60 or renal dysfunction)
- **Ifosfamide:** mesna with every dose; watch for encephalopathy
- **HBV:** entecavir if HBsAg+ or anti-HBc+ on rituximab · G-CSF with every cycle · quinolone during neutropenia

## R-HYPER-CVAD / R-MA (THOMAS, CANCER 2006) — ALTERNATING, 6–8 CYCLES

| | Dosing |
|---|---|
| **A (1,3,5,7)** | Rituximab 375 · cyclophosphamide 300 q12h ×6 d1–3 (mesna) · doxorubicin 50 CIV d4 · vincristine 2 mg d4+11 · dexamethasone 40 mg d1–4, 11–14 |
| **B (2,4,6,8)** | Rituximab 375 · **MTX 200 over 2 h + 800 over 22 h d1** · **cytarabine 3000 q12h ×4 d2–3** (1000 if >60 yr or Cr >1.5) · leucovorin from 36 h |
| **CNS** | IT MTX 12 mg d2 + IT cytarabine 100 mg d8, cycles 1–4 (8 IT). CNS+: twice weekly until CSF clear ×2 |

## HIGH-DOSE MTX — URINE ALKALINIZATION

| | |
|---|---|
| **Pre-check** | **Drain or avoid if effusion / ascites** (third-space reservoir) · stop **PPIs, TMP-SMX, penicillins, NSAIDs, salicylates, probenecid** · CrCl: 50% dose if 10–50, omit <10 |
| **Alkalinize** | ½NS + 100 mEq $NaHCO_3$/L at 500 mL/h ×2 h, then 125 mL/h. **Start MTX only when urine pH ≥7** and UOP >100 mL/h; keep pH ≥7 until cleared |
| **Levels** | q24h from 24 h. **Escalate leucovorin to 100 mg/m² q6h** if at 24 h Cr ↑>50% or level >20 µM; 48 h >1 µM; 72 h >0.1 µM |
| **Glucarpidase** | **50 U/kg IV** if level >1 µM with delayed clearance from renal failure. **No leucovorin within 2 h** either side; **immunoassay unreliable for 48 h** (use chromatographic assay) |

Contrast with TLS (page 1): urine alkalinization is **required for HD-MTX** (MTX precipitates in acid urine) but **harmful for TLS**. When both overlap (cycle 1 d10), rasburicase has already handled urate — alkalinize for the MTX.

## KEY TRIALS AND COHORTS

| | |
|---|---|
| **LY10** *Mead, Blood 2008* | Dose-modified CODOX-M/IVAC (MTX 3 g/m²), prospective: 2-yr PFS 64% in BL; low risk 85%, high risk 49% |
| **LMB ± R** *Ribrag, Lancet 2016* | Phase 3, n=260, HIV–: rituximab 3-yr EFS **75% vs 62%**, HR 0.59 — the rituximab randomization |
| **NCI 9177** *Roschewski, JCO 2020* | Risk-adapted DA-EPOCH-R, single arm: EFS 84.5%; **CSF+ is the failure group** |
| **HOVON/SAKK** *Chamuleau, Lancet Haem 2023* | R-CODOX-M/R-IVAC vs DA-EPOCH-R, CNS– high risk: similar PFS, fewer infections with EPOCH |
| **MDACC R-hyper-CVAD** *Thomas, Cancer 2006* | n=31: 3-yr OS 89%; rituximab improved outcome vs historical hyper-CVAD |
| **BL-IPI** *Olszewski, JCO 2021* | Age ≥40, ECOG ≥2, LDH >3×, CNS — 3-yr PFS 92 / 72 / 53%; validated in 457 ex-US patients |
| **US real-world** *Evens, Blood 2021* | 641 adults, 30 centres: 3-yr PFS 64%, OS 70%, **TRM 10%**; no difference by HIV status |
| **CNS cohort** *Zayac, Haematologica 2021* | CNS+ in 19%; CR 59% vs 77%. 3-yr CNS relapse **13% after DA-EPOCH-R** vs 4% CODOX-M/IVAC, 3% hyper-CVAD/MA |
| **HIV-BL** *Alderuccio, Blood Adv 2021* | n=249 US-UK: 3-yr PFS 61%; CD4 did not predict; CNS relapse higher after DA-EPOCH |

## PEARLS

- **Board triad: MYC-IG translocation + BCL2-negative + Ki-67 ~100%.** BCL2-positive with MYC = think double hit, not Burkitt
- **Urine alkalinization: never for TLS, always for HD-MTX**
- Don't wait for complete staging to start TLS prophylaxis — several pediatric-derived protocols (LMB) open with a low-intensity **pre-phase** (cyclophosphamide, vincristine, prednisone) to debulk
- **If CSF is positive, don't choose DA-EPOCH-R**; if the patient is old, HIV+ or frail and CSF-negative, DA-EPOCH-R is the kinder, equally effective choice
- No maintenance, no consolidative transplant in CR1